

BALAJI RADHAKRISHNAN PADMANABHAN

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SUMMARY

Graduate student pursuing a Master of Science in Computer Software Engineering at Arizona State University with 3 years of Quality Assurance experience supporting enterprise healthcare applications. Experienced in SQL data validation, database testing, ETL and data flow validation, data reconciliation, and root cause analysis of data quality issues across business critical systems. Currently building hands on data engineering and analytics skills through Python, Pandas, PostgreSQL, SQL, and Tableau, with a Bachelor's specialization in AI and Machine Learning. Interested in developing AI enabled analytics tools, improving enterprise data quality, and resolving data ownership and source of truth challenges.

EDUCATION

Master of Science, Computer Software Engineering, GPA: 3.56

Aug 2025 – May 2027

Arizona State University, Ira A. Fulton Schools of Engineering

Tempe, AZ

Bachelor of Technology, Computer Science & Engineering (AI and ML)

Jun 2018 – May 2022

SRM Institute of Science and Technology

Chennai, India

TECHNICAL SKILLS

Programming & Querying: Python, SQL

Data Engineering: ETL Pipelines, Data Integration, Data Cleaning, Data Normalization, Data Transformation, Data Modeling, Data Validation, Data Quality, Data Reconciliation, Schema Validation

Data Processing: Pandas, NumPy, CSV, JSON

Analytics & Visualization: Tableau, Microsoft Excel, Microsoft PowerPoint

AI & GenAI: Retrieval-Augmented Generation (RAG), OpenAI API, Google Gemini API, LLM Prompt Design

Databases & Analytics Tools: PostgreSQL, MySQL, dbt, Snowflake

Development & Collaboration Tools: Git, GitHub, Docker, GitHub Actions, Jenkins, Postman, JIRA, Zephyr, Confluence, ServiceNow, BMC Remedy

Testing & Methodologies: Database Testing, API Testing, Integration Testing, Regression Testing, UAT, Agile Scrum, SDLC, Root Cause Analysis, Process Documentation

PROFESSIONAL EXPERIENCE

Tata Consultancy Services – Quality Assurance Analyst

Sept 2022 – July 2025

Client: Cigna Healthcare

Chennai, India

- Partnered with Product, Engineering, and QA teams across 50+ Scrum sprints as part of a 12-member Agile team supporting Cigna's claims and member-portal applications.
- Designed and maintained 300+ test cases in JIRA and Zephyr covering API, database, functional, integration, regression, UAT, smoke, and end-to-end testing across release cycles.
- Wrote SQL queries using PostgreSQL and MySQL to validate data integrity, reconcile datasets, verify business rules, compare expected and actual records, and investigate data discrepancies.
- Performed end-to-end database and data-flow validation across interconnected application layers to verify the completeness, consistency, and accurate processing of business-critical healthcare data.
- Used Postman to validate REST API request and response payloads, JSON structures, HTTP status codes, required fields, business rules, and error-handling scenarios.
- Developed and executed Python scripts to automate repetitive regression checks and routine data-validation activities, improving testing efficiency and consistency.
- Identified and tracked 150+ defects through resolution, documented reproducible evidence, performed root cause analysis of recurring data issues, and revalidated fixes to prevent recurrence.
- Validated application builds through Jenkins and GitHub Actions pipelines before UAT sign-off and production release approval.
- Created Excel-based defect reports and trend analyses while maintaining test results, data-validation evidence, and process documentation in Confluence for distributed India and US teams.
- Mentored 2 junior QA testers on test execution, defect reporting, SQL-based data validation, and Agile QA practices.

Arizona State University – MIX Center

March 2026 – Present

IT Deskside Support Assistant (Student Worker)

Mesa, AZ

- Triage and resolve daily walk-in and remote ServiceNow tickets covering hardware, software, AV, and account-access issues while documenting reproduction steps, troubleshooting activities, and resolution notes.
- Maintain asset inventory records in ServiceNow and BMC Remedy with close attention to data accuracy, record consistency, and source-of-truth alignment across systems of record.
- Reconcile device and asset information between platforms, identify mismatched and conflicting records, and update documentation to maintain reliable inventory data.
- Reimage and configure lab and end-user devices through SCCM using PXE boot, WinPE task sequences, and post-deployment verification while supporting 10+ tickets per week.

ACADEMIC PROJECTS & PUBLICATIONS

Global Superstore ELT Pipeline with AI Chat Interface | *Python, Pandas, PostgreSQL, SQL, OpenAI API, Tableau, Git* Summer 2026

- Built an end-to-end ELT pipeline consolidating, cleaning, and normalizing 51,000+ retail sales records into a structured, maintainable data model: raw ingestion layer in PostgreSQL, in-database SQL transformation executed by a transactional Python runner, and a one-command orchestrator that runs the full pipeline and exports BI-ready datasets.
- Diagnosed fully corrupted date columns (all 51,290 values destroyed by spreadsheet save damage) through source-data profiling, and recovered week-level order dates by deriving them from surviving year and week-number fields after cross-validating against order-ID-embedded years (zero mismatches).
- Documented six data-quality findings, each with reproducible SQL evidence queries and an explicit finding-evidence-decision narrative, covering corrupted fields, constant junk columns, and redundant groupings.
- Implemented idempotent transformations (DROP ... CASCADE with CREATE TABLE AS SELECT in a single transaction) so the full pipeline reruns safely from scratch in one command.
- Developed a retrieval-augmented generation (RAG) chatbot in Python using the OpenAI GPT API that answers natural-language questions grounded in the semantic SQL views, shows source attribution for every answer, and declines questions the retrieved data cannot support.
- Designed semantic SQL views (regional performance, monthly trends via the recovered dates, product profitability with exception flags) as the governed consumption layer for the RAG chatbot and a Tableau dashboard (in development).

AI-Ready Analytics Pipeline with dbt and Snowflake (Workshop Project) | *dbt, Snowflake, SQL, Git* Summer 2026

- Completed a hands-on workshop building a production-style analytics workflow that transformed raw data into tested and documented analytical models using dbt and Snowflake.
- Applied modular SQL transformations and data-modeling practices to organize datasets for reliable downstream analytics and reporting.
- Implemented dbt data-quality tests and model documentation to validate assumptions and improve trust in transformed datasets.
- Explored how governed semantic models can support reliable natural-language analytics and AI-driven data applications, including retrieval-based question answering over structured data.

Software Requirements and Specification: Data-Driven Planning Poker | *CSE 563*

Fall 2025

- Collaborated on a requirements engineering study proposing a data-driven alternative to traditional Planning Poker using historical software effort information.
- Authored stakeholder analysis, customer problems supported by literature, operational storyboards, and IEEE 29148-aligned functional and non-functional requirements.

Publication: *“Designing A Dynamic Framework For People Counting Using YOLO-PC”*

Mar 2024

- Presented at ICCCN 2022, Manchester Metropolitan University; published in AIP Conference Proceedings, March 2024.